

Class 7

Land, Rails, and the National Market

Part II — Building the American Growth Machine

The American Economy — draft course materials (proposal under discussion)

The Golden Spike: Promontory Summit, Utah, May 10, 1869



Source: Andrew J. Russell, "East and West Shaking Hands at the Laying of the Last Rail" (1869). Public domain, via Wikimedia Commons. The transcontinental railroad was completed in Utah.

The point of today's class

America became rich partly because it built networks — railroads, roads, ports, and broadband — that turned thousands of separated local economies into one enormous integrated market.

Why it matters

- ▶ If you grow wheat in Kansas, make steel in Pittsburgh, or write code in Utah — who is your market?
- ▶ Not just who *wants* your product: who you can *reach* cheaply enough.
- ▶ For most of history, distance dominated economic life:
 - ▶ wagons, canals, and rivers moved almost everything
 - ▶ early cities hugged the water because water was cheap transport
- ▶ Markets do not just exist — they are **built**. That photo from Utah is the national market being built.

Today

1. Land first: turning the frontier into property
2. Railroads build the national market
3. Market access and the counterfactual (Donaldson–Hornbeck)
4. What the network did to what America produces
5. Highways: the second national network
6. Broadband: the third

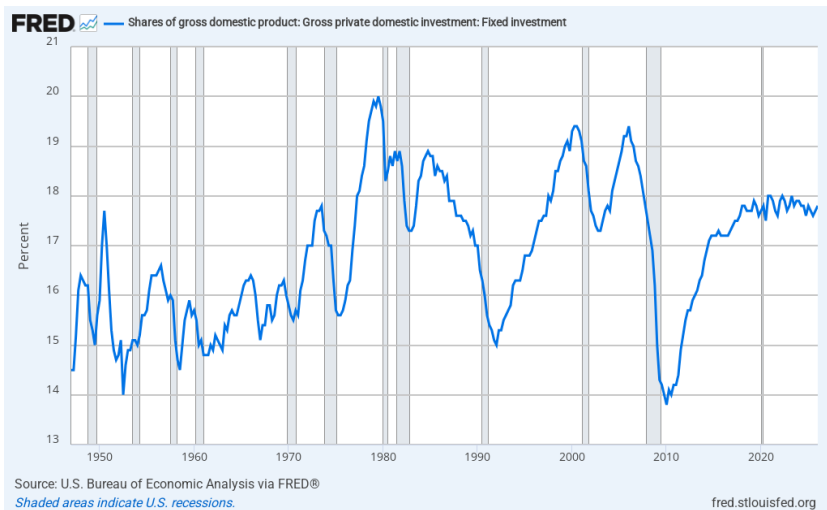
Land first: turning the frontier into property

- ▶ **Rectangular survey**: the grid made land legible — standardized, titled, tradable plots across a continent.
- ▶ **Land offices**: recorded and sold title — formal property rights at continental scale.
- ▶ **Homestead Act (1862)**: 160 acres in exchange for settlement — mass smallholding, not a landed aristocracy.
- ▶ Secure, tradable title is the *first* market infrastructure: collateral for loans, incentive to invest, easy to sell and move on.
- ▶ This is where the transformation from roughly 80% of workers on farms to under 2% begins.

Railroads build the national market

- ▶ Between 1850 and 1890 the rail network explodes — the U.S. becomes *internally* connected.
- ▶ Not just more capital — a **network**: it grows more valuable as more places connect.
- ▶ Huge **fixed costs**: a railroad pays off only at scale.
- ▶ **General equilibrium**: a railroad changes prices, land values, production, migration, and competitors *everywhere* — not just in one town.

U.S. fixed private investment as a share of GDP



Source: Bureau of Economic Analysis, via FRED (A007RE1Q156NBEA).

Wealth-to-income ratios, 1700–present

& non-profit sectors).

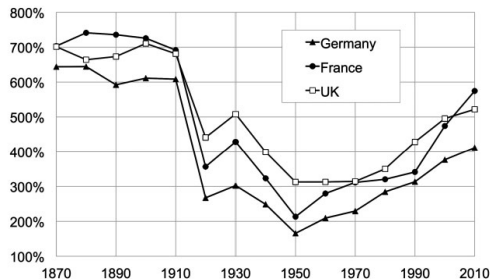


FIGURE II

Private Wealth-National Income Ratios in Europe, 1870–2010

Authors' computations using country national accounts. Private wealth = non-financial assets + financial assets – financial liabilities (household & non-profit sectors). Data are decennial averages (1910–1913 averages for 1910).

Source: Piketty and Zucman (2014), *Quarterly Journal of Economics*, "Capital is Back."

Market access: the Donaldson–Hornbeck question

- ▶ Wrong question: “does my county have a railroad?” Right question: how much does the network improve my access to *all* markets?
- ▶ Market access is **capitalized into farmland values** — land prices summarize expected future profits.
- ▶ The counterfactual: remove the railroads and re-route trade over wagons and canals — agricultural land values collapse.
- ▶ Why “rail counties are richer” misleads: railroads were not placed randomly. The course-wide question: **compared with what?**

The network changes what America produces

- ▶ **Regional specialization:** the wheat belt, the cotton belt, the industrial Northeast.
- ▶ **Chicago:** rail + grain + meatpacking + Great Lakes + finance — the hub-and-spoke city.
- ▶ **Agglomeration:** dense, connected places become more productive.
- ▶ **Path dependence:** once the network shapes where people and firms locate, the effects persist for generations.

Highways: the second national network

- ▶ The interstate system builds the 20th-century national market: trucks, cars, suburbs, logistics.
- ▶ Baum-Snow's question: *did highways cause suburbanization?* Central-city vs. metro population, 1950–1990.
- ▶ Trucking, container ports, distribution centers, fulfillment — modern retail is an infrastructure story.

Broadband: market access without physical shipment

- ▶ Broadband is market access for jobs, education, telehealth, finance, and remote work.
- ▶ Rural gaps: is lacking broadband today like lacking rail access in 1890?
- ▶ Remote work partly detaches opportunity from geography — but agglomeration still matters.

Railroads, highways, broadband — the same story three times

19th century	20th century	21st century
Railroads	Highways	Broadband, cloud, logistics
Grain, livestock, manufactures	Cars, suburbs, trucking	E-commerce, remote work
Market towns and rail hubs	Suburbs and distribution centers	Platforms and data centers
Land values repriced	City-suburb patterns repriced	Housing and job access repriced

- ▶ Every network creates **total gains and winners and losers**: bypassed towns, neighborhoods divided by highways, rural areas left off the broadband map.

Discussion

Railroads, highways, and broadband all reduce the cost of distance. Pick one: how did it change market access? Who gained, who lost — and what is the relevant counterfactual?

Read before next class

- ▶ Donaldson and Hornbeck (2016), “Railroads and American Economic Growth: A Market Access Approach,” *QJE* (skim the maps).
- ▶ Baum-Snow (2007), “Did Highways Cause Suburbanization?” *QJE* (skim the figures).